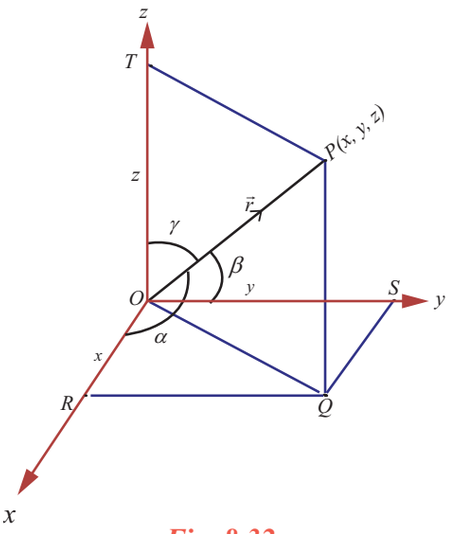


Fig. 8.32

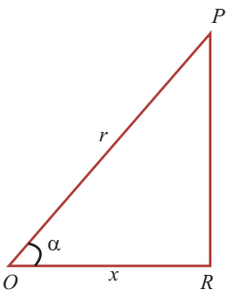


Fig. 8.33

In $\Delta OPR, \angle PRO = 90°, \angle POR = \alpha, OR = x,$ and $OP = r.$ Therefore

$$\cos\alpha = \frac{OR}{OP} = \frac{x}{r}.$$

In a similar way we can find that $\cos\beta = \frac{y}{r}$ and $\cos\gamma = \frac{z}{r}$.

Here the angles α, β, γ are called direction angles of the vector $\overrightarrow{OP} = \vec{r}$ and $\cos\alpha, \cos\beta, \cos\gamma$ are called **direction cosines** of the vector $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$. Thus $\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$, where $r = \sqrt{x^2 + y^2 + z^2}$, are the direction cosines of the vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

Any three numbers which are proportional to the direction cosines of vector are called the **direction ratios** of the vector. Hence the direction ratios of a vector is not unique. For a given vector, we have, infinitely many set of direction ratios.

Observations

(i) For a given non-zero vector, one can find the direction ratios as well as the direction cosines.
(ii) For a given set of direction ratios, one cannot find the corresponding vector.
(iii) For a given set of direction cosines, one cannot find the corresponding vector.
(iv) For a given vector, the triplet of direction cosines is also a triplet of direction ratios.
(v) To find the vector, the magnitude as well as either the set of direction cosines or a set of direction ratios are essential.

Note 8.4

Here we consider a vector $\overrightarrow{OP}$ whose initial point is at the origin. If the vector whose initial point is not the origin, then, in order to find its direction cosines, we draw a vector with initial point at the origin and parallel to the given vector of same magnitude by translation. By the principle of two equal vectors having the same set of direction cosines, we can find direction cosines of any vector.

Result 8.11

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be the position vector of any point and let α, β, γ be the direction angles of $\vec{r}$. Then

(i) the sum of the squares of the direction cosines of $\vec{r}$ is 1.

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